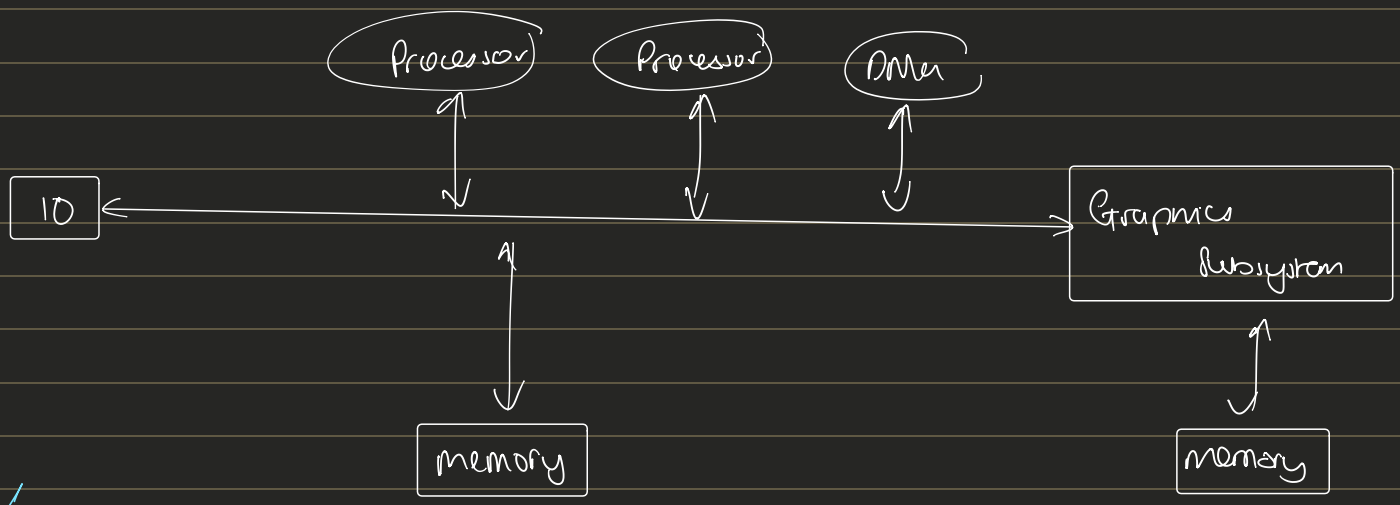
Lecture 2 : 06/02/26

Buses

Bus is an 'overloaded' term

↳ We'll use bus to compass the logical bundle of stuff between processors.

Privelege modes needed for certain accesses.

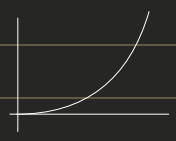


↳ System architecture is quite complex, it can have many devices.

For buses we can have interconnection. → Possible mixture of shared and switch buses.

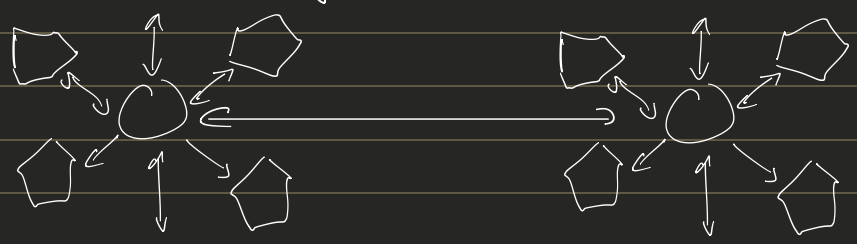
↳ Single Buses : Simple, easy. Bottlenecks.

Crossbar buses : Offers Parallelism Scales badly $O(n^2)$.



We often have NoC, network on chip.

→ often has a rectangular grid pattern (suits silicon layout)



Other topologies ARE available (irregular ones).

Bytes and Words:

Byte usually is the smallest addressable unit
→ 8 bits

The size of 'word' varies.
This is usually the 'natural' data size in the architecture.
Increasingly 32 bits (4 bytes)
↳ Half word (16 bits), doubleword 64 bits etc.

Alignment:

Bytes have particular widths: 32, 64 etc.

Misaligned operations will be slower (or malfunction)
→ You want them to be aligned.
↓
If you don't align them and let it occur you do use all memory slots but it will be slow as the data will be across 2 words. meaning you will need 2 memory reads.

In memory space is important but performance more so.

Code alignment may be enforced on some architectures
- RISC (fixed length instructions).

Some ISAs have variable length opr.
- x86
- Alignment is inflexible

Endianness - How the bytes are arranged in a word.
↳ Big Endian, Little Endian.

Most machines default to little endian.
More complex machines can swap their bases to better endian.

Von Neumann vs Harvard.

Most mainstream processors are Von Neumann.
↳ at least at the software view. not always architectural

A common approach:
- A Harvard interface to the processor, A Von Neumann interface for the programmer.

System Bus Organisation
↓
You have a hierarchy of memory and buses.